

Convergence in probability and Relative entropy method for moderate interacting system

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Setting and literature

Stochastic moderate interacting system

Model with N-particle moderate interacting system

$$dX_i^{N,\varepsilon} = \frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon(X_i^{N,\varepsilon} - X_j^{N,\varepsilon}) dt + \sqrt{2} dW_t^i$$

$X_i^N(0) = \xi_i \in \mathbb{R}^d$, i.i.d, ρ_0 is the probability density function of ξ_i .

where $\tilde{\Phi}_\varepsilon$ is smoothed version of given singular interaction potential Φ .

Example 2-D Keller-Segel system. $\Phi(x) = -\frac{1}{2\pi} \ln |x|$, $\tilde{\Phi}_\varepsilon$ can be given by

$$\tilde{\Phi} = \begin{cases} -\frac{1}{2\pi} \ln |x|, & |x| > 2\varepsilon, \\ -\frac{1}{4\pi\varepsilon} |x| + \frac{1}{2\pi} (1 - \ln 2\varepsilon), & |x| \leq 2\varepsilon, \end{cases}, \quad \tilde{\Phi}_\varepsilon = j_\varepsilon * \tilde{\Phi}.$$

The corresponding ε -McKean-Vlasov equation (fix ε and $N \rightarrow \infty$) is

$$d\bar{X}_i^\varepsilon = \nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon) dt + \sqrt{2} dW_t^i.$$

where the law of $\bar{X}_i^\varepsilon$ satisfies

$$\partial_t \rho^\varepsilon - \Delta \rho^\varepsilon = \nabla \cdot (\rho^\varepsilon \nabla c^\varepsilon), \quad c^\varepsilon = \tilde{\Phi}_\varepsilon * \rho^\varepsilon,$$

Expected mean-field limit with scalling $\varepsilon(N) \rightarrow 0$ as $N \rightarrow \infty$

McKean-Vlasov equation

$$d\hat{X}_i = \nabla \Phi * \rho(\hat{X}_i) dt + \sqrt{2} dW_t^i.$$

where the law of $\hat{X}_i$ satisfies

$$\partial_t \rho - \Delta \rho = \nabla \cdot (\rho \nabla c), \quad c = \Phi * \rho,$$

Benefit from PDE solutions:

Compare to the classical work in mean-field limit (measure valued solutions).
 $\nabla \Phi * \rho$ can be bounded Lipschitz for “good” ρ .

Remark: It is known that log scaling $\varepsilon \sim 1/\log N$ is “almost” trivial (proof similar to the bounded Lipschitz case). Reason: “too much” cut-off.

Goal: Algebraic scaling $\varepsilon = N^{-\beta}$ for some $\beta > 0$.

Selected Literature Review

Mean field limit (for non-Lipschitz potentials)

- K. Oelschläger (1987): repulsive δ , L^2 estimate with rate $o(N^{-1/2})$
- Lazarovici and Pickl (2017): Vlasov, Convergence in probability
- Jabin and Wang, (2018): relative entropy
- S. Serfaty (2020): Coulomb potential, modulated energy
- Rosenzweig and Serfaty (2023): Reisz potential
- Olivera, Richard, and Tomasevic (2023): Semigroup approach
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Main result and proof ideas

Consider the high dimensional PDEs with factorized initial data $\rho_0^{\otimes N}$.

$$\partial_t \rho^{N,\varepsilon} - \sum_{i=1}^N \Delta_{x_i} \rho^{N,\varepsilon} + \sum_{i=1}^N \nabla_{x_i} \left(\rho^{N,\varepsilon} \frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon(x_i - x_j) \right) = 0$$

$$\partial_t \rho^{\otimes N} - \sum_{i=1}^N \Delta_{x_i} \rho^{\otimes N} + \sum_{i=1}^N \nabla_{x_i} \left(\rho^{\otimes N} \nabla \Phi * \rho(x_i) \right) = 0$$

where $\rho^{N,\varepsilon}$ is the joint law of $X_i^{N,\varepsilon}$, $1 \leq i \leq N$.

k - marginal of $\rho^{N,\varepsilon}$, $1 \leq k \leq N$ is given by

$$\rho^{N,\varepsilon;k}(x_{k+1}, \dots, x_N) = \int_{\mathbb{R}^{d(N-k)}} \rho^{N,\varepsilon}(x_1, \dots, x_N) dx_{k+1} \cdots dx_N.$$

Theorem (Mean field limit)

Let $\rho^{N,\varepsilon;1}$ be the 1-particle marginal of $\rho^{N,\varepsilon}$ and ρ be the solution of the limiting PDE, then $\forall \beta \in (0, \beta_1)$ ($\beta_1 = \frac{1}{8}$ for K-S), it holds

$$\|\rho^{N,\varepsilon;1} - \rho\|_{L^1(\mathbb{R}^d)} \leq \frac{C(t)}{N^\beta}.$$

Idea:

Convergence in probability “+” Relative entropy
(Originally: Lazarovici and Pickl) (Originally: Jabin and Wang)

New: Mean field limit in L^1 norm for moderately interacting system with Coulomb potential on $\mathbb{R}^d$.

Key Step: Use convergence in probability to close the relative entropy estimate for moderate interacting systems.

Convergence in probability

Proposition (Convergence in probability)

For any $\alpha \in (0, \frac{1}{2})$ satisfies $0 < \beta < \min\{\frac{\alpha}{3}, \frac{1}{4}, \frac{1}{2} - \alpha\}$, then $\forall \gamma > 0$ it holds

$$\sup_{0 \leq t \leq T} \mathbb{P} \left(\max_{1 \leq i \leq N} |X_i^{N,\varepsilon} - \bar{X}_i^\varepsilon| \geq N^{-\alpha} \right) \leq \frac{C(\gamma)}{N^\gamma}.$$

Idea: Stopping time, law of large numbers (Lazarovici and Pickl)

A key tool: A version of law of large numbers

Lemma

let $(\bar{Y}_i)_{i=1, \dots, N}$ be i.i.d random variables, $\nu \in L^1(\mathbb{R}^2)$ be their density, $U \in L^\infty(\mathbb{R}^2)$ be given, Then for any $\tilde{k} \geq 1$, we have:

$$\mathbb{P}(\mathcal{A}_\theta^N(U, \nu)) \leq N \max_{1 \leq i \leq N} \mathbb{P}(\mathcal{A}_\theta^i(U, \nu)) \leq N^{2\tilde{k}(\theta - \frac{1}{2})+1} C(\tilde{k}) \|U\|_{L^\infty}^{2\tilde{k}}.$$

where $\mathcal{A}_\theta^N(U, \nu) = \cap_{i=1}^N \mathcal{A}_\theta^i(U, \nu)$ with

$$\mathcal{A}_\theta^i(U, \nu) = \left\{ \omega \in \Omega : \left| \frac{1}{N} \sum_{j=1}^N U(\bar{Y}^i - \bar{Y}^j) - U * \nu(Y^i) \right| > \frac{1}{N^\theta} \right\}.$$

Estimate for the cut-offed potential

Lemma (2-D K-S)

$$\|\nabla^2 \tilde{\Phi} * g\|_{L^\infty} \leq C(\|g\|_{L^1} - \ln 2\varepsilon \|g\|_{L^\infty}).$$

Stopping time

Let $X_i^{N,\varepsilon}$ and $\bar{X}_i^\varepsilon$ be the solution of particle system and the ε - McKean Vlasov equation. We define the stopping time $\tau(\omega)$ by

$$\tau(\omega) = \inf\{t \in (0, T] \mid \max_{1 \leq i \leq N} |X_i^{N,\varepsilon}(t) - \bar{X}_i^\varepsilon(t)| \geq N^{-\alpha}\}.$$

and introduce the cut-off process $S(\omega, t)$

$$S(\omega, t) = N^{2\alpha k} \max_{1 \leq i \leq N} |(X_i^{N,\varepsilon} - \bar{X}_i^\varepsilon)(t \wedge \tau)|^{2k} \leq 1.$$

where $k \in \mathbb{N}$ is to be determined later. Markov's inequality implies that

$$\sup_{0 \leq t \leq T} \mathbb{P}\left(\max_{1 \leq i \leq N} |X_i^{N,\varepsilon} - \bar{X}_i^\varepsilon| \geq N^{-\alpha}\right) \leq \sup_{0 \leq t \leq T} \mathbb{P}(S_t = 1) \leq \sup_{0 \leq t \leq T} \mathbb{E}(S_t).$$

Evolution of $S(t)$

Take the difference, apply Ito's formula, and take the expectation,

$$\begin{aligned} \mathbb{E}(S(t)) &= N^{2\alpha k} \mathbb{E} \left(\max_{1 \leq i \leq N} |(X_i^{N,\varepsilon} - \bar{X}_i^\varepsilon)(t \wedge \tau(\omega))|^{2k} \right) \\ &\leq CN^\alpha \mathbb{E} \left(\max_{1 \leq i \leq N} \int_0^{t \wedge \tau} |\nabla \tilde{\Phi}_\varepsilon * \mu_N(X_i^{N,\varepsilon}(s)) - \nabla \tilde{\Phi}_\varepsilon * \bar{\mu}_N(\bar{X}_i^\varepsilon(s))| S(s)^{\frac{2k-1}{2k}} ds \right) \\ &\quad + CN^\alpha \mathbb{E} \left(\max_{1 \leq i \leq N} \int_0^{t \wedge \tau} |\nabla \tilde{\Phi}_\varepsilon * \bar{\mu}_N(\bar{X}_i^\varepsilon(s)) - \nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon(s))| S(s)^{\frac{2k-1}{2k}} ds \right) \\ &:= K_1 + K_2, \end{aligned}$$

where

$$\mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{X_j^{N,\varepsilon}(s)}, \quad \bar{\mu}_N = \frac{1}{N} \sum_{j=1}^N \delta_{\bar{X}_j^\varepsilon(s)}$$

are the empirical measures.

Estimate for K_2 , law of large numbers Let

$$A_{\theta_1}(\nabla \tilde{\Phi}_\varepsilon, \rho^\varepsilon) = \cup_{i=1}^N A_{\theta_1}^i,$$

and obtain the estimate for K_2

$$\begin{aligned} K_2 &\leq CN^\alpha \mathbb{E} \left(\max_{1 \leq i \leq N} \int_0^{t \wedge \tau} |\nabla \tilde{\Phi}_\varepsilon * \bar{\mu}_N(\bar{X}_i^\varepsilon(s)) - \nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon(s))| S(s)^{\frac{2k-1}{2k}} ds \right) \\ &\leq CN^{2\alpha k} \mathbb{E} \left(\max_i \int_0^{t \wedge \tau} (\mathbb{I}_{A_{\theta_1}^c} + \mathbb{I}_{A_{\theta_1}}) \left(\frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon * \bar{\mu}_N - \nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon \right)^{2k} (\bar{X}_i^\varepsilon) ds \right) \\ &\quad + C \int_0^t \mathbb{E}(S(s)) ds \\ &\leq CN^{2\alpha k} \left(\frac{1}{N^{2\theta_1 k}} + \|\nabla \tilde{\Phi}_\varepsilon\|_{L^\infty}^{2k} \mathbb{P}(A_{\theta_1}) \right) + C \int_0^t \mathbb{E}(S(s)) ds \\ &\leq C(N^{(\alpha-\theta_1)2k} + N^{2\alpha k} C(\tilde{k}) \|\nabla \tilde{\Phi}_\varepsilon\|_{L^\infty}^{2(k+\tilde{k})} N^{2\tilde{k}(\theta_1 - \frac{1}{2})+1}) + \int_0^t \mathbb{E}(S(s)) ds. \end{aligned}$$

Estimate for K_1

By using Taylor's expansion at point $\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon$ we have

$$\begin{aligned}
 K_1 &\leq CN^\alpha \mathbb{E} \left(\max_i \int_0^{t \wedge \tau} \left| \frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon(X_i^{N,\varepsilon} - X_j^{N,\varepsilon}) - \nabla \tilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon) \right| S^{\frac{2k-1}{2k}} ds \right) \\
 &\leq CN^\alpha \mathbb{E} \left(\max_i \int_0^{t \wedge \tau} \left| \frac{1}{N} \sum_{j=1}^N \nabla^2 \tilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)(X_i^{N,\varepsilon} - X_j^{N,\varepsilon} - (\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)) \right| S^{\frac{2k-1}{2k}} ds \right) \\
 &\quad + CN^\alpha \mathbb{E} \left(\max_i \int_0^{t \wedge \tau} \|\nabla^3 \tilde{\Phi}_\varepsilon\|_{L^\infty} \left| \frac{1}{N} \sum_{j=1}^N (X_i^{N,\varepsilon} - X_j^{N,\varepsilon} - (\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)) \right|^2 S^{\frac{2k-1}{2k}} ds \right) \\
 &:= K_{11} + K_{12}.
 \end{aligned}$$

where K_{12} can be estimated by

$$K_{12} \leq \frac{C}{N^\alpha} \|\nabla^3 \tilde{\Phi}_\varepsilon\|_{L^\infty} \int_0^t \mathbb{E}(S(s)) ds.$$

Estimate for K_{11}

For K_{11} , we need to use the law of large numbers

$$\begin{aligned}
 K_{11} &\leq CN^\alpha \mathbb{E} \left(\max_i \int_0^{t \wedge \tau} \frac{1}{N} \sum_{j=1}^N |\nabla^2 \tilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)| \max_i |X_i^{N,\varepsilon} - \bar{X}_i^\varepsilon| S^{\frac{2k-1}{2k}} ds \right) \\
 &\leq C \mathbb{E} \left(\max_i \int_0^{t \wedge \tau} \frac{1}{N} \sum_{j=1}^N (|\nabla^2 \tilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)| - |\nabla^2 \tilde{\Phi}_\varepsilon| * \rho^\varepsilon(\bar{X}_i^\varepsilon)) S ds \right) \\
 &\quad + C \mathbb{E} \left(\max_i \int_0^{t \wedge \tau} (|\nabla^2 \tilde{\Phi}_\varepsilon| * \rho^\varepsilon(\bar{X}_i^\varepsilon)) S ds \right) \\
 &=: K_{111} + K_{112},
 \end{aligned}$$

where

$$K_{112} \leq C \| |\nabla^2 \tilde{\Phi}_\varepsilon| * \rho^\varepsilon \|_{L^\infty} \int_0^t \mathbb{E}(S(s)) ds \leq C(1 - \ln 2\varepsilon) \int_0^t \mathbb{E}(S(s)) ds.$$

Estimate for K_{111} , law of large numbers

Take $U = |\nabla^2 \tilde{\Phi}_\varepsilon|$, $\nu = \rho^\varepsilon$ and

$$A_\theta(|\nabla^2 \tilde{\Phi}_\varepsilon|, \rho^\varepsilon) = \cup_{i=1}^N A_\theta^i \Rightarrow A_\theta^c(|\nabla^2 \tilde{\Phi}_\varepsilon|, \rho^\varepsilon) = \cap_{i=1}^N (A_\theta^i)^c.$$

then with $\theta = 0$ we have

$$\begin{aligned} K_{111} &\leq C \mathbb{E} \left(\max_i \int_0^{t \wedge \tau} \frac{1}{N} \sum_{j=1}^N (|\nabla^2 \tilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)| - |\nabla^2 \tilde{\Phi}_\varepsilon| * \rho^\varepsilon(\bar{X}_i^\varepsilon)) S ds \right) \\ &\leq C \mathbb{E} \left((\mathbb{I}_{A_\theta^c} + \mathbb{I}_{A_\theta}) \max_i \int_0^{t \wedge \tau} \left(\frac{1}{N} \sum_{j=1}^N |\nabla^2 \tilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)| - |\nabla^2 \tilde{\Phi}_\varepsilon| * \rho^\varepsilon(\bar{X}_i^\varepsilon) \right) S ds \right) \\ &\leq \frac{C}{N^\theta} \int_0^t \mathbb{E}(S(s)) ds + C \|\nabla^2 \tilde{\Phi}_\varepsilon\|_{L^\infty} \mathbb{P}(A_\theta) \\ &\leq C \int_0^t \mathbb{E}(S(s)) ds + C \|\nabla^2 \tilde{\Phi}_\varepsilon\|_{L^\infty} \|\nabla^2 \tilde{\Phi}_\varepsilon\|_{L^\infty}^{2\tilde{k}} N^{2\tilde{k}(\theta - \frac{1}{2}) + 1}. \end{aligned}$$

Gronwall's inequality

Notice for 2-D K-S, $\|\nabla^3 \tilde{\Phi}_\varepsilon\|_{L^\infty} \leq \varepsilon^{-3}$, $\|\nabla^2 \tilde{\Phi}_\varepsilon\|_{L^\infty} \leq \varepsilon^{-2}$, $\|\nabla \tilde{\Phi}_\varepsilon\|_{L^\infty} \leq \varepsilon^{-1}$.
 we have

$$\begin{aligned} \mathbb{E}(S(t)) &\leq C \int_0^t \left(1 + N^{-\alpha} \|\nabla^3 \tilde{\Phi}_\varepsilon\|_{L^\infty} - \ln 2\varepsilon\right) \mathbb{E}(S(s)) ds + CN^{(\alpha-\theta_1)2k} \\ &\quad + CN^{2\tilde{k}(\theta-\frac{1}{2})+1} \|\nabla^2 \tilde{\Phi}_\varepsilon\|_{L^\infty}^{2\tilde{k}+1} + CN^{2\alpha k+2\tilde{k}(\theta_1-\frac{1}{2})+1} \|\nabla \tilde{\Phi}_\varepsilon\|_{L^\infty}^{2(k+\tilde{k})} \\ &\leq C \int_0^t \left(1 + N^{3\beta-\alpha} + \ln N^\beta\right) \mathbb{E}(S(s)) ds + CN^{(\alpha-\theta_1)2k} \\ &\quad + CN^{2\tilde{k}(-\frac{1}{2}+2\beta)+(2\beta+1)} + CN^{2k(\alpha+\beta)+2\tilde{k}(\theta_1-\frac{1}{2}+\beta)+1}. \end{aligned}$$

Therefore for $0 < \alpha < \frac{1}{2}$, we take $0 < \beta < \min\{\frac{\alpha}{3}, \frac{1}{4}, \frac{1}{2} - \alpha\}$ and $\theta_1 \in (\alpha, \frac{1}{2} - \beta)$. Then $\forall \gamma > 0$, there exists k and $\tilde{k}$ s.t.

$$\mathbb{E}(S(t)) \leq CN^{-2\gamma} + C \int_0^t \left(1 + N^{3\beta-\alpha} + \ln N^\beta\right) \mathbb{E}(S(s)) ds.$$

Relative entropy

A key tool: Relative entropy method Consider the high dimensional PDEs with factorized initial data $\rho_0^{\otimes N}$.

$$\partial_t \rho^{N,\varepsilon} - \sum_{i=1}^N \Delta_{x_i} \rho^{N,\varepsilon} + \sum_{i=1}^N \nabla_{x_i} \left(\rho^{N,\varepsilon} \left(\frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon(x_i - x_j) - \nabla \tilde{\Phi}_\varepsilon * \tilde{f}_N(x_i) \right) \right) = 0$$

$$\partial_t \rho^{\otimes N} - \sum_{i=1}^N \Delta_{x_i} \rho^{\otimes N} + \sum_{i=1}^N \nabla_{x_i} \left(\rho^{\otimes N} (\nabla \Phi * \rho(x_i) - \nabla \Phi * \tilde{f}_N f(x_i)) \right) = 0$$

The relative entropy of N-particle system $\mathcal{H}$ defined by

$$\mathcal{H}(\rho^{N,\varepsilon} | \rho^{\otimes N}) = \frac{1}{N} \mathcal{H}_N(\rho^{N,\varepsilon} | \rho^{\otimes N}) = \frac{1}{N} \int_{\mathbb{R}^{2N}} \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \cdot \rho^{\otimes N} dX_N.$$

Proposition (Relative entropy estimate (Using convergence in probability))

If $\varepsilon = N^{-\beta}$, $0 < \beta < \beta_1 (= \frac{1}{8})$ for 2-D KS, then

$$\mathcal{H}(\rho^{N,\varepsilon} | \rho^{\otimes N}) \leq \frac{C(t)}{N^{2\beta}},$$

Compute the time evolution of the relative entropy, we have

$$\begin{aligned}
 & \frac{d}{dt} \mathcal{H}(\rho^{N,\varepsilon} | \rho^{\otimes N}) + \frac{1}{N} \int_{\mathbb{R}^{2N}} \sum_{i=1}^N \left| \nabla_{x_i} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \right|^2 \rho^{N,\varepsilon} dX_N \\
 &= \int_{\mathbb{R}^{2N}} \frac{1}{N} \sum_{i=1}^N \left(\nabla \Phi * \rho(x_i) - \frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon(x_i - x_j) \right) \nabla_{x_i} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \rho^{N,\varepsilon} dX_N \\
 &\leq \frac{1}{2N} \int_{\mathbb{R}^{2N}} \sum_{i=1}^N \left| \nabla_{x_i} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \right|^2 \rho^{N,\varepsilon} dX_N \\
 &\quad + \frac{1}{2} \int_{\mathbb{R}^{2N}} \frac{1}{N} \sum_{i=1}^N \left| \nabla \Phi * \rho(x_i) - \frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon(x_i - x_j) \right|^2 \rho^{N,\varepsilon} dX_N.
 \end{aligned}$$

As a summary,

$$\begin{aligned}
 & \frac{d}{dt} \mathcal{H}(\rho^{N,\varepsilon} | \rho^{\otimes N}) + \frac{1}{2N} \int_{\mathbb{R}^{2N}} \sum_{i=1}^N \left| \nabla_{x_i} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \right|^2 \rho^{N,\varepsilon} dX_N \\
 &\leq \frac{1}{2} \mathbb{E} \left(\frac{1}{N} \sum_{i=1}^N \left| \nabla \Phi * \rho(X_i^{N,\varepsilon}) - \frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon(X_i^{N,\varepsilon} - X_j^{N,\varepsilon}) \right|^2 \right) := M.
 \end{aligned}$$

Use convergence in probability to close the estimate of M

$$\begin{aligned}
 2M &\leq \mathbb{E} \left(\frac{1}{N} \sum_i |\nabla \Phi * \rho(X_i^{N,\varepsilon}) - \nabla \Phi_\varepsilon * \rho(X_i^{N,\varepsilon})|^2 \right) \leq \varepsilon^2 (1 + \|\rho\|_{W^{1,q}}^2) \\
 &+ \mathbb{E} \left(\frac{1}{N} \sum_i |\nabla \Phi_\varepsilon * \rho(X_i^{N,\varepsilon}) - \nabla \tilde{\Phi}_\varepsilon * \rho(X_i^{N,\varepsilon})|^2 \right) \leq \varepsilon^2 (1 + \|\rho\|_{W^{1,q}}^2) \\
 &+ \mathbb{E} \left(\frac{1}{N} \sum_i |\nabla \tilde{\Phi}_\varepsilon * \rho(X_i^{N,\varepsilon}) - \nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon(X_i^{N,\varepsilon})|^2 \right) \leq \varepsilon^2 (1 + \|\rho\|_{W^{1,q}}^2) \\
 &+ \mathbb{E} \left(\frac{1}{N} \sum_i |\nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon(X_i^{N,\varepsilon}) - \nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon)|^2 \right) \\
 &+ \mathbb{E} \left(\frac{1}{N} \sum_i |\nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon) - \frac{1}{N} \sum_{j=1}^N \nabla \tilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)|^2 \right) \\
 &+ \mathbb{E} \left(\frac{1}{N} \sum_i \left| \frac{1}{N} \sum_{j=1}^N (\nabla \tilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon) - \nabla \tilde{\Phi}_\varepsilon(X_i^{N,\varepsilon} - X_j^{N,\varepsilon})) \right|^2 \right) \\
 &:= \sum_{l=1}^6 M_l.
 \end{aligned}$$

Since $\bar{X}_i^\varepsilon$ are i.i.d. random variables, we know

$$M_5 \leq \frac{1}{N} \|\nabla \tilde{\Phi}_\varepsilon\|_{L^\infty}^2 \leq N^{2\beta-1}. \quad \text{law of large numbers}$$

By the result of convergence in probability, we have for set

$$\mathcal{A}_\alpha = \{\omega \in \Omega \mid \max_{1 \leq i \leq N} |X_i^{N,\varepsilon} - \bar{X}_i^\varepsilon| > N^{-\alpha}\},$$

the estimate

$$\mathbb{P}(\mathcal{A}_\alpha) \leq \frac{C(\gamma)}{N^\gamma}.$$

Therefore, $M_4 + M_6$

$$\begin{aligned} &\leq C(\gamma) \frac{\|\nabla \tilde{\Phi}_\varepsilon\|_{L^\infty}^2 + \|\nabla \tilde{\Phi}_\varepsilon * \rho^\varepsilon\|_{L^\infty}^2}{N^\gamma} + \frac{\|D^2 \tilde{\Phi}_\varepsilon\|_{L^\infty}^2 + \|D^2 \tilde{\Phi}_\varepsilon * \rho^\varepsilon\|_{L^\infty}^2}{N^{2\alpha}} \\ &\leq C(\gamma) \frac{\varepsilon^2}{N^\gamma} + \frac{\varepsilon^4}{N^{2\alpha}} \leq C(N^{2\beta-\gamma} + N^{4\beta-2\alpha}) \leq CN^{-\frac{2\alpha}{3}}. \end{aligned}$$

The L^1 convergence follows from:

- Relative entropy estimate:

$$\mathcal{H}(\rho^{N,\varepsilon} | \rho^{\otimes N}) \leq C \min\{N^{-\beta}, N^{-\frac{2\alpha}{3}}\} \rightarrow 0.$$

- Super-Additivity of Relative Entropy:

$$\mathcal{H}(\rho^{N,\varepsilon} | \rho^{\otimes N}) = \frac{1}{N} \mathcal{H}_N(\rho^{N,\varepsilon} | \rho^{\otimes N}) \geq \frac{1}{2k} \mathcal{H}_k(\rho^{N,\varepsilon;k} | \rho^{\otimes k})$$

- Csiszàr-Kullback-Pinsker:

$$\|f - g\|_{L^1(\mathbb{R}^d)}^2 \leq 2\mathcal{H}(f|g)$$

Therefore, we obtain

$$\|\rho^{N,\varepsilon;1} - \rho\|_{L^1(\mathbb{R}^d)} \rightarrow 0.$$

Possible generalizations

- Second order system (Vlasov Poisson, Cucker-Smale)
- Convergence in probability for weaker norms
- Fermionic quantum systems
- Further fluctuation estimates
- Derivation of cross-diffusion systems
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Happy Birthday, Shih-Hsien !